

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 09 · Basic Trigonometry

With Answers

Questions with full worked solutions

- 20 pages
- Past papers up to June 2026

Basic Trigonometry

Your classified route

7
questions

42
marks

3
families

5
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Basic Trigonometry

Families and variants

F01 • Area, angle and side linkage

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

V02 Side ratio gives largest angle; area fixes the scale

F02 • Sine rule and the ambiguous case

V01 SSA sine rule with longest-side branch selection

F03 • Multi-step triangle synthesis

V01 Derive a side relation, solve it exactly, then find exact area

V02 Composite triangles: derive cosine expression, then propagate with sine rule and area

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Area, angle and side linkage

WHAT THIS FAMILY ASKS

Move between area, included angle and side length, making the correct acute/obtuse choice before applying the cosine rule.

VARIANT ROUTE

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

V02 Side ratio gives largest angle; area fixes the scale

4 Complete Classified questions

21 total marks

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

(a) the size of angle DAB , in degrees, to 2 decimal places, (3)

(b) the length of diagonal BD , in cm , to one decimal place. (2)

(Total 5 marks)

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Question 1

WMA11/01 October 2019

T09 Basic Trigonometry

F01 Area, angle and side linkage

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

1

Use the area of a parallelogram**Area** $= ab \sin \theta$, so $40 = 10 \cdot 6 \sin \theta$. Hence $\sin \theta = \frac{2}{3}$.

2

Choose the obtuse angle $\theta = 180^\circ - \sin^{-1}\left(\frac{2}{3}\right) = 138.19^\circ$ to 2 decimal places.

3

Use the cosine rule for BD $BD^2 = 10^2 + 6^2 - 2(10)(6) \cos(138.19^\circ)$. Therefore $BD \approx 15.014 \dots$, so $BD = 15.0$ cm to 1 decimal place.**FINAL ANSWER**(a) $\angle DAB = 138.19^\circ$. (b) $BD = 15.0$ cm.

2. The triangle ABC is such that

- $AB = 15$ cm
- $AC = 25$ cm
- angle $BAC = \theta^\circ$
- area triangle $ABC = 100 \text{ cm}^2$

(a) Find the value of $\sin \theta^\circ$

(2)

Given that $\theta > 90$

(b) find the length of BC , in cm, to 3 significant figures.

(3)

(Total for Question 2 is 5 marks)

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Question 2

WMA11/01 January 2024

T09 Basic Trigonometry

F01 Area, angle and side linkage

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

1

Use the triangle area formula

$$100 = \frac{1}{2}(15)(25) \sin \theta.$$

2

Find sine theta

$$\sin \theta = \frac{200}{375} = \frac{8}{15}.$$

3

Use theta greater than 90 degrees

Since $\theta > 90^\circ$, $\cos \theta$ is negative. So $\cos \theta = -\sqrt{1 - \left(\frac{8}{15}\right)^2} = -\frac{\sqrt{161}}{15}.$

4

Apply the cosine rule

$$BC^2 = 15^2 + 25^2 - 2(15)(25) \cos \theta = 850 + 50\sqrt{161}.$$

5

Round the length

$$BC = \sqrt{850 + 50\sqrt{161}} \approx 38.5 \text{ cm to 3 significant figures.}$$

FINAL ANSWER

$$(a) \sin \theta = \frac{8}{15}. \quad (b) BC = 38.5 \text{ cm.}$$

3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15 \text{ m}$
- $AC = 10 \text{ m}$
- angle $BAC = \theta^\circ$

(a) show that $\sin \theta^\circ = \frac{4}{5}$ (1)

Given also that θ° is the **largest** angle in the triangle,

(b) find the length of the side BC , in metres, to 3 significant figures, (3)

(c) find the size of the **smallest** angle in the triangle, in degrees, to one decimal place. (2)

(Total for Question 3 is 6 marks)

Question 3

WMA11/01 May/June 2026

T09 Basic Trigonometry

F01 Area, angle and side linkage

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

1

(a) Use the area formula

$$60 = \frac{1}{2}(15)(10) \sin \theta^\circ = 75 \sin \theta^\circ, \text{ so } \sin \theta^\circ = \frac{60}{75} = \frac{4}{5}.$$

2

(b) Select the angle consistent with the information

$$\sin^{-1}\left(\frac{4}{5}\right) = 53.130\dots^\circ, \text{ but } \theta \text{ is the largest angle, so it must be the obtuse}$$

$$\text{alternative: } \theta = 180^\circ - 53.130\dots^\circ = 126.869\dots^\circ. \text{ Hence } \cos \theta = -\frac{3}{5}.$$

3

(b) Apply the cosine rule

$$BC^2 = 15^2 + 10^2 - 2(15)(10) \cos \theta = 225 + 100 + 180 = 505. \text{ Therefore } BC = \sqrt{505} = 22.472\dots \text{ m} = 22.5 \text{ m to 3 significant figures.}$$

4

(c) Identify the smallest angle

The shortest side is $AC = 10$, so the smallest angle is $\angle ABC$.

5

(c) Use the sine rule

$$\frac{\sin \angle ABC}{10} = \frac{\sin \theta}{\sqrt{505}}, \text{ so } \sin \angle ABC = \frac{8}{\sqrt{505}}. \text{ Thus } \angle ABC = 20.854\dots^\circ = 20.9^\circ \text{ to one decimal place.}$$

FINAL ANSWER

$$(a) \sin \theta^\circ = \frac{4}{5}. \quad (b) BC = 22.5 \text{ m}. \quad (c) \angle ABC = 20.9^\circ.$$

3. The sides of a triangle are in the ratio 2:4:5

(a) Find the size of the **largest** angle of the triangle, in degrees, to one decimal place. (2)

Given that the area of the triangle is 140 cm^2

(b) find the length of the **shortest** side, in cm, to one decimal place. (3)

(Total for Question 3 is 5 marks)

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Question 4

WMA11/01 October 2025

T09 Basic Trigonometry

F01 Area, angle and side linkage

V02 Side ratio gives largest angle; area fixes the scale

1

Use the side ratio

Take the side lengths as $2x$, $4x$ and $5x$. The largest angle is opposite the side $5x$.

2

Use the cosine rule

$$(5x)^2 = (2x)^2 + (4x)^2 - 2(2x)(4x) \cos \theta.$$

3

Find the angle

$$25 = 4 + 16 - 16 \cos \theta, \text{ so } \cos \theta = -\frac{5}{16} \text{ and } \theta = 108.2^\circ \text{ to one decimal place.}$$

4

Use the area formula

$$140 = \frac{1}{2}(2x)(4x) \sin 108.2^\circ = 4x^2 \sin 108.2^\circ.$$

5

Find the scale factor

$$x^2 = \frac{140}{4 \sin 108.2^\circ}, \text{ so } x \approx 6.070.$$

6

Shortest side

The shortest side is $2x \approx 12.1$ cm.

FINAL ANSWER

(a) 108.2° . (b) **12.1 cm.**

Sine rule and the ambiguous case

WHAT THIS FAMILY ASKS

Use an SSA configuration to calculate a sine value and decide between supplementary angles from geometric information.

VARIANT ROUTE

V01 SSA sine rule with longest-side branch selection

1 Complete Classified questions

5 total marks

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places. (2)

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places. (3)

(Total 5 marks)

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Question 5

WMA11/01 May/June 2022

T09 Basic Trigonometry

F02 Sine rule and the ambiguous case

V01 SSA sine rule with longest-side branch selection

1

Use the sine rule

The side opposite x is $AB = 21$, and the side opposite 25° is $BC = 13$, so

$$\frac{\sin x}{21} = \frac{\sin 25^\circ}{13}.$$

2

Find the sine value

$$\sin x = \frac{21 \sin 25^\circ}{13} = 0.68269 \dots$$

3

Choose the correct angle

The angle from the calculator is 43.05° , but AB is the longest side, so x is obtuse. Hence $x = 180^\circ - 43.05^\circ = 136.95^\circ$.

FINAL ANSWER

$x = 137^\circ$ to 3 significant figures.

Multi-step triangle synthesis

WHAT THIS FAMILY ASKS

Chain cosine rule, algebra, sine rule and area across one or more triangles while preserving exact values as long as possible.

VARIANT ROUTE

V01 Derive a side relation, solve it exactly, then find exact area

V02 Composite triangles: derive cosine expression, then propagate with sine rule and area

2 Complete Classified questions

16 total marks

3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

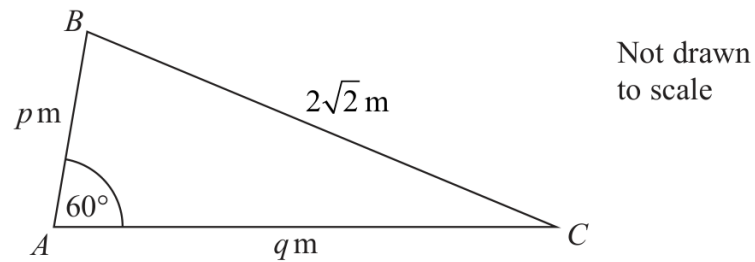


Figure 1

Figure 1 shows the plan view of a flower bed.

The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

(a) Show that

$$p^2 + q^2 - pq = 8 \quad (2)$$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
- (ii) the exact value of q . (5)

Using the answers to part (b),

- (c) calculate the exact area of the flower bed. (2)

(Total 9 marks)

Question 6

WMA11/01 May/June 2021

T09 Basic Trigonometry

F03 Multi-step
synthesis

triangle

V01 Derive a side relation,
solve it exactly, then find exact
area

1

Apply the cosine rule

$BC^2 = AB^2 + AC^2 - 2(AB)(AC) \cos 60^\circ$. Since $BC = 2\sqrt{2}$, $AB = p$ and $AC = q$, this gives $8 = p^2 + q^2 - 2pq \left(\frac{1}{2}\right)$.

2

Show the required result

Therefore $p^2 + q^2 - pq = 8$.

3

Use $q = p + 2$

Side AC is 2 metres longer than AB , so $q = p + 2$. Substitute into $p^2 + q^2 - pq = 8$.

4

Solve for p

$p^2 + (p + 2)^2 - p(p + 2) = 8$, so $p^2 + 2p - 4 = 0$. Hence $p = -1 \pm \sqrt{5}$. Since $p > 0$, $p = \sqrt{5} - 1$.

5

Find q

$q = p + 2 = \sqrt{5} + 1$.

6

Find the exact area

Area = $\frac{1}{2}pq \sin 60^\circ$. Now $pq = (\sqrt{5} - 1)(\sqrt{5} + 1) = 4$, so area = $\frac{1}{2}(4) \left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} \text{ m}^2$.

FINAL ANSWER

(a) $p^2 + q^2 - pq = 8$. (b) $p = \sqrt{5} - 1$, $q = \sqrt{5} + 1$. (c) $\sqrt{3} \text{ m}^2$.

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5.

Diagram **NOT**
accurately drawn

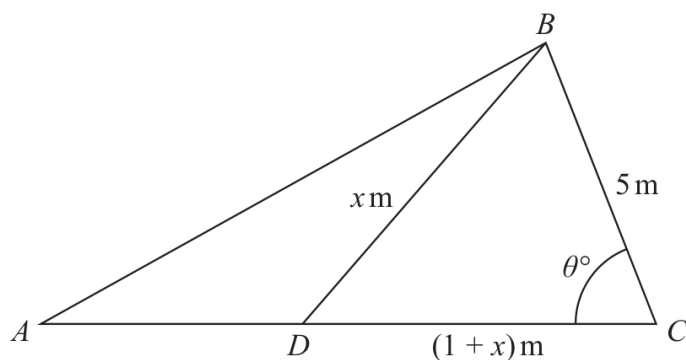


Figure 2

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

- $BD = x \text{ m}$
- $CD = (1 + x) \text{ m}$
- $BC = 5 \text{ m}$
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30^\circ$
- ADC is a straight line

(b) find the area of triangle ABC , giving your answer, in m^2 , to one decimal place.

(5)

(Total for Question 5 is 7 marks)

Question 7

WMA11/01 October 2023

T09 Basic Trigonometry

F03 Multi-step
synthesis

triangle

V02 Composite triangles: derive cosine expression, then propagate with sine rule and area

1

Apply the cosine rule in triangle BCD

$$x^2 = 5^2 + (1+x)^2 - 2(5)(1+x) \cos \theta.$$

2

Rearrange for cos theta

$$10(1+x) \cos \theta = 25 + (1+x)^2 - x^2 = 26 + 2x, \text{ so } \cos \theta = \frac{26+2x}{10(1+x)} = \frac{13+x}{5+5x}.$$

3

Substitute x equals 2 root 3

$$\cos C = \frac{13+2\sqrt{3}}{5+10\sqrt{3}}, \text{ so } C \approx 42.47^\circ.$$

4

Use angle sum

$$A = 30^\circ, \text{ so } B = 180^\circ - 30^\circ - C \approx 107.53^\circ.$$

5

Use the sine rule for the area

$$\text{Since } BC = 5 \text{ and } A = 30^\circ, \frac{BC}{\sin A} = 10. \text{ Hence } AB = 10 \sin C \text{ and } AC = 10 \sin B.$$

6

Calculate the area

$$\text{Area} = \frac{1}{2}(AB)(AC) \sin A = 50 \sin C \sin B \approx 32.2 \text{ m}^2.$$

FINAL ANSWER

$$(a) \cos \theta = \frac{13+x}{5+5x}. \quad (b) \text{ Area} = 32.2 \text{ m}^2.$$